AI-BASED ATTENDANCE MONITORING SYSTEM

Automating Attendance with Artificial Intelligence, Computer Vision & Face Recognition

Introduction

The AI-Based Attendance Monitoring System is an intelligent software solution developed to automate the process of recording and managing attendance. Traditional methods, such as calling names, signing registers, or manually entering records, are time-consuming and prone to errors. This project overcomes these limitations by using Artificial Intelligence, Computer Vision, and Face Recognition technology.

The system captures faces through a camera, detects the presence of individuals, compares their facial features with previously registered data, and automatically marks their attendance. Attendance information — including the person's identity, date, and time — is stored digitally for future reference, reducing manual effort and improving the accuracy and efficiency of attendance management.

GitHub Repository

Repository Name: AI-Based-Attendance-Monitoring-System

Repository Link: github.com/pooja21-chaturvedi/AI-Based-Attendance-Monitoring-System

Version Control Operations Demonstrated

• Initialized the local repository with git init and linked it to the remote GitHub repository using git remote add origin <link>.

• Staged and committed project files using git add . and git commit -m "message".

• Pushed committed changes to GitHub using git push origin main.

• Cloned the repository using git clone <repository link> to verify the setup on another machine.

Project Modules

Module 1: System Development & User Registration

Assigned To: Pooja Chaturvedi (Developer)

Designs the overall system architecture and builds the user registration feature, capturing name, ID, and facial data used later for recognition.

Module 2: AI & Face Recognition

Assigned To: Parul Goel (Expert)

Implements face detection and recognition, comparing captured faces with registered data to accurately identify users.

Module 3: Attendance Management & Improvement

Assigned To: Prabal Pratap (Adviser)

Plans the workflow from recognition to attendance storage, prevents duplicate entries, and guides reporting improvements.

Module 4: Testing & Validation

Assigned To: Prakhar (Tester)

Tests registration, detection, recognition, and attendance marking to ensure the system is accurate and reliable.

Expected Outcome

A functional AI-based system that automatically identifies registered users and records their attendance digitally, reducing manual effort while improving the accuracy and efficiency of attendance management.

Project Team

#	Name	Roll No.	Role
1	Pooja Chaturvedi	202401100500125	Developer
2	Parul Goel	202401100500124	Expert
3	Prabal Pratap	202401100500126	Adviser
4	Prakhar	202401100500128	Tester